CODEALPHA CYBERSECURITY INTERNSHIP | TASK 2

Phishing Awareness Training

Spot it. Stop it. Report it.

Recognize

Resist

Report

Presented by: Your Name

AGENDA

What we will cover

1. **Why it matters**
   The numbers behind phishing
2. **How phishing works**
   The attack lifecycle and types
3. **Recognize it**
   Emails, websites, texts and calls
4. **Social engineering**
   The psychology of the trick
5. **Real cases**
   Famous attacks and lessons
6. **Defend and respond**
   Best practices, quiz and takeaways

WHY IT MATTERS

Why phishing matters

68%

of breaches involved a human element

Source: Verizon DBIR 2024

$2.9B

lost to business email compromise

Source: FBI IC3 Report 2023

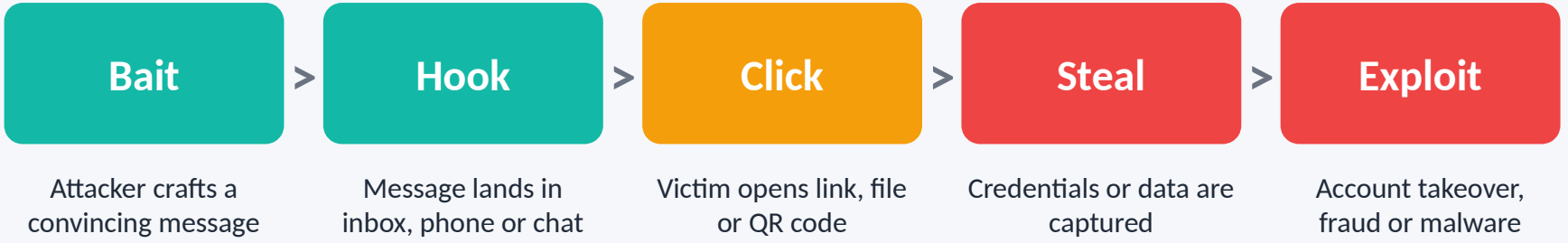
#1

most reported cybercrime type: phishing

Source: FBI IC3 Report 2023

Attackers go after people because people are easier to fool than firewalls.

How a phishing attack works



Break the chain early

The earlier you stop the attack, the less damage it does. Spotting the bait and not clicking is the cheapest defense; reporting it quickly protects everyone else.

Types of phishing

Email phishing
Mass emails posing as banks, couriers or IT

Spear phishing
Personalised message aimed at one person

Whaling
Targets executives and senior leaders

Smishing
Fake SMS with malicious links

Vishing
Phone calls pretending to be bank or support

Quishing
Malicious QR codes leading to fake logins

RECOGNIZE IT

Anatomy of a phishing email

Inbox > New message

From: PayPal Support <support@paypa1-secure.com>
Subject: URGENT: Account suspended in 24 hours!

Dear customer,

We noticed unusal activity on your acount. Verify now or your account will be closed.

[Verify My Account] http://paypa1-secure.com/login

Attachment: invoice_2026.zip

1. **Look-alike sender**
   paypa1 uses the number 1, not the letter l
2. **Urgency and threats**
   Pressure stops you from thinking
3. **Generic greeting**
   Real companies use your name
4. **Spelling mistakes**
   Unusal, acount
5. **Suspicious link**
   Hover first to see the real URL
6. **Risky attachment**
   Unexpected .zip, .exe or macro files

RECOGNIZE IT

Spotting fake websites

Real vs fake web addresses

REAL https://www.paypal.com

Correct domain

FAKE https://paypa1.com

Number 1 replaces l

FAKE https://paypal.com.secure-login.xyz

Real domain is secure-login.xyz

FAKE http://paypal-verify-account.net

Extra words, no HTTPS

Website checklist

Read the domain letter by letter.
The real domain is the last two parts before the first single slash.
Be wary of pages asking for passwords or card details out of the blue.
A padlock only means the connection is encrypted. Phishing sites can have one too.
Type the address or use a bookmark.

RECOGNIZE IT

Smishing and vishing in the wild

Text message (unknown number)

DHL: Your parcel could not be delivered. Pay a 1.99 fee to reschedule: http://dhl-redelivery.top/pay

Red flags: unexpected delivery, small fee, odd domain (.top), no tracking number.

Vishing call script

Hello, this is your bank's fraud team. We detected a suspicious transfer. For your safety, read us the code we just sent to your phone.

What to do

Hang up. Call the number on your card or the official site. Real banks never ask for one-time codes or PINs.

Social engineering tactics

Urgency
Act now or lose access

Fear
Threats of fines, arrest or closure

Authority
Posing as the CEO, IT, bank or police

Greed and curiosity
Prizes, refunds, 'see this photo'

Pretexting
An invented story to gain trust

Baiting and tailgating
Infected USB drives or following staff through doors

Real-world cases

Google and Facebook (2013-15)
A fraudster posed as a hardware vendor and invoiced both firms for over 100 million dollars.

Twitter (2020)
Attackers phoned staff, obtained credentials and took over high-profile accounts.

Target (2013)
Credentials stolen from a phished HVAC vendor led to a major card-data breach.

Business Email Compromise
Fake CEO or supplier emails ask for urgent wire transfers or new bank details.

DEFEND

The S.T.O.P. check



Sender

Does the address match exactly?



Tone

Urgent, scary or too good to be true?



Ask

Does it want passwords, money or secrecy?



Path

Hover links. Check the domain. Skip surprise files.

Any red flag? Don't click. Report it.

DEFEND

Best practices for individuals

Pause and verify
Use a separate, trusted channel

Don't click or open
Skip unexpected links and files

Use MFA
A stolen password alone won't work

Password manager
Won't autofill on fake domains

Update software
Patches close attacker doors

Report, don't reply
Forward to IT, then delete

DEFEND

Defenses organizations use

Email filtering
Spam filters and attachment sandboxing

SPF, DKIM and DMARC
Stop attackers spoofing your domain

Phishing-resistant MFA
Passkeys and hardware security keys

Simulated phishing
Regular tests that train staff

One-click report button
Fast reporting to the security team

Incident playbook
A clear plan for when someone clicks

RESPOND

Clicked a suspicious link? Act fast

1. Disconnect from the network
2. Change passwords, starting with the affected account
3. Tell IT or security straight away
4. Turn on MFA and review account activity
5. Run a malware scan and watch for unusual charges

QUIZ

Spot the phish

Email A

From: IT Helpdesk <helpdesk@micros0ft-support.com>
Subject: Password expires TODAY

Dear user, click here to keep your account: http://bit.ly/reset-now

Failure to act will lock you out.

Email B

From: Priya Shah <priya.shah@yourcompany.com>
Subject: Agenda for Thursday's meeting

Hi Sam, attached is the agenda we discussed in this morning's stand-up. Let me know if you'd like any changes.

Which one is phishing, and what gave it away? Vote now.

QUIZ

Quiz time

1. Email from it-support@micros0ft-help.com demands a reset within 1 hour. Red flag?
2. True or false: a padlock means a website is safe.
3. Your 'boss' texts: buy gift cards now and keep it quiet. What do you do?
4. Safest link? (a) paypal.com.secure-login.net (b) paypal.com (c) paypa1.com
5. You clicked a phishing link. First move?

QUIZ

Quiz answers

1. Look-alike domain (zero in micros0ft) plus artificial urgency.
2. False. Phishing sites can use HTTPS too.
3. Verify by calling them directly. Do not buy anything. Report it.
4. (b) paypal.com
5. Disconnect, change passwords and tell IT right away.

Key takeaways

1. Slow down. Urgency is the attacker's best tool.
2. Check the sender and the real domain.
3. Verify through a separate, trusted channel.
4. Use MFA and a password manager.
5. When in doubt, report it.

NEXT STEPS

Resources and thank you

CISA
cisa.gov: phishing guidance and alerts

Anti-Phishing Working Group
apwg.org: phishing trends and reporting

Have I Been Pwned
haveibeenpwned.com: check if your email was in a breach

Thank you! Questions?
Source code, quiz and this presentation:
github.com/<your-username>/CodeAlpha_PhishingAwarenessTraining